

CENG381 Stochastic Processes

Hitting Times — Practice 1

Q1 (30 points). Consider the Markov chain on $\{0, 1, 2, 3, 4\}$ where states 0 and 4 are absorbing, and each interior state k moves left or right with probability $1/2$ each. Let $h_k = P(\text{absorbed at } 4 \mid \text{start } k)$ and $f_k = E[\text{absorption time} \mid \text{start } k]$.

- Write the recurrence and boundary conditions for h_k . Solve to find h_1, h_2, h_3 .
 - Write the recurrence and boundary conditions for f_k . Use $f_k = k(n-k)$ with $n=4$ to find f_1, f_2, f_3 . Verify f_2 by direct substitution into the recurrence.
 - Starting from state 2, conditional on eventually being absorbed at state 4, what is the probability the chain reaches state 3 before state 1? (Hint: this is the same as the conditional absorption probability.)
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Q2 (35 points). A fair coin is flipped repeatedly. Let T be the first time the sequence HH (two consecutive heads) appears. Model the process as a Markov chain on states:

State 0 (Start): no relevant suffix yet
State 1 (H): the most recent flip was H
State 2 (HH): the sequence HH just appeared (absorbing)

- Write the 3×3 transition matrix P . (From State 0: go to State 1 with prob $1/2$, stay at 0 with prob $1/2$. From State 1: go to State 2 with prob $1/2$, go back to State 0 with prob $1/2$.)
 - Let $\tau_0 = E[T \mid \text{start at State 0}]$ and $\tau_1 = E[T \mid \text{start at State 1}]$. Write the system of two equations for τ_0 and τ_1 .
 - Solve for τ_0 . Compare your answer with the expected time to see HTH, which equals 10. Why is HH faster to see than HTH?
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Q3 (35 points). A Markov chain on $\{A, B, C, D\}$ has transition matrix:

	A	B	C	D
A	[1/2	1/2	0	0]
B	[1/4	1/2	1/4	0]
C	[0	1/4	1/2	1/4]
D	[0	0	0	1]

State D is absorbing. Let $\tau_x = E[\text{first time to reach D} \mid \text{start at state } x]$.

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- a) Write the system of equations for τ_A , τ_B , τ_C . ($\tau_D = 0$ by convention.)
- b) Solve the system. (Hint: first eliminate τ_A by expressing it in terms of τ_B , then substitute.)
- c) Explain why $\tau_A = \infty$. Can state D ever be reached starting from A? What does this imply about the classification of state A?